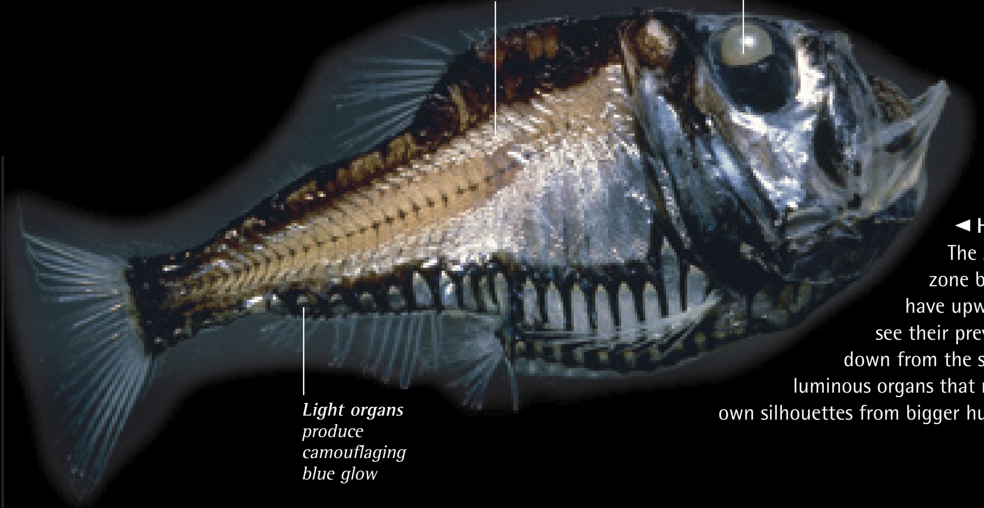
▲ VERTICAL MIGRATIONS
Zooplankton such as these copepods graze on phytoplankton near the surface at night, but at dawn they retreat to the twilight zone up to 3,300 ft (1,000 m) below, to avoid shoals of hungry fish. They migrate upward again in the evening—a long journey for such tiny creatures.

THE OCEAN DEPTHS

In the dark waters beneath the sunlit zone, the primary food source is dead material that drifts down from above. This has low food value compared to the living phytoplankton near the surface, so the zooplankton that live in the twilight zone by day migrate toward the surface at night to feed. Fish that eat the zooplankton migrate upward with them, and are targeted from below by bigger predators. Many of these fish glow in the dark, and have huge mouths and long teeth for seizing scarce prey. By contrast, most of the animals that live on the deep ocean floor feed on dead animals and other debris.



ocean depths

◄ HATCHETFISH
The zooplankton that live in the twilight zone by day are hunted by hatchetfish. These have upward-facing eyes that enable them to see their prey against the dim blue light filtering down from the surface. Their bellies are studded with luminous organs that match the blue glow, concealing their own silhouettes from bigger hunters.

► WEIRD GLOW
Thousands of deep-water animals like this squid glow in the dark. They have light organs called photophores, which use a chemical reaction to generate light without heat. The animals use the light to confuse their enemies, to signal to one another in the gloom, and to lure or spotlight their prey.

◄ MONSTERS OF THE DEEP
Prey is so scarce in the depths that hunters must be able to eat anything they run into. The monstrous-looking gulper eel can swallow animals as big as itself, thanks to its enormous mouth and an elastic stomach that can stretch like a balloon.